STAT 236 Final Project

Sparse Submatrix Detection

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1 Introduction

Sparse submatrix detection is a problem in high-dimensional statistics concerned with detecting a small, structured signal hidden within a large noisy matrix. As with many of the topics covered in this course, a central question is how the size of the planted submatrix and the strength of its signal affect our ability to distinguish it from random noise.

An important feature of this problem is the distinction between statistical and computational limits of detection. From a purely statistical perspective, we would like to determine the weakest signal for which any hypothesis test can reliably distinguish the two cases, regardless of its computational cost. However, a statistically optimal procedure may require searching over an enormous number of possible submatrices. We therefore also ask how strong the signal must be for detection to be possible using an efficient algorithm, such as one running in polynomial time.

The problem was studied by Butucea and Ingster (2013) [1], who established asymptotic detection boundaries for the Gaussian model, and later by Ma and Wu (2015) [2], who investigated the computational limitations of detection. Our hope is to develop the theory in an accessible way, so that an undergraduate or graduate student can follow the main arguments and gain an understanding of the basic literature. To this end, we primarily consider square matrices with square planted submatrices and a common positive mean shift.

The papers contain substantial technical detail, and working through every argument in full generality would be beyond what we could reasonably understand within the scope of this project. We explain the main detection guarantees in the text and put the calculations in the appendix. We then compare the detection regimes and discuss the stronger requirements for recovering the affected rows and columns.

Table 1 summarizes the procedures discussed in this report. Here, n is the number of rows and columns in the observed matrix, k is the number of affected rows and columns, and μ is their common mean shift. The noise entries are independent standard Gaussian variables. We assume $k \to \infty$ and $k = o(n)$. For comparisons with polynomial block growth, we write $k = \lfloor n^\gamma \rfloor$, where $0 < \gamma < 1$.

Table 1. Detection and recovery procedures

Procedure	**Paper**	**Goal**	**Regime and guarantee**	**Cost**
Sum	Butucea and Ingster [1]	Detection	Statistically rate optimal for $\gamma > 2/3$. Reliable when $\mu k^2/n \to \infty$.	$O(n^2)$
Scan	Butucea and Ingster [1]	Detection	Attains the statistical exponent boundary for $\gamma < 2/3$. The sufficient signal condition is given in Section 2.3.	Exhaustive search
Maximum entry	Ma and Wu [2]	Detection	A sufficient regime is $\liminf \mu/(2\sqrt{\log n}) > 1$, where individual entries carry strong signals.	$O(n^2)$
Sum and scan	Butucea and Ingster [1]	Detection	Combines the two detection regimes and attains the statistical boundary.	Includes exhaustive scan
Maximizing block sum	Butucea, Ingster and Suslina [3]	Recovery	Attains the sharp statistical boundary for exact recovery stated in Section 4.1.	Exhaustive search
Spectral projection	Cai, Liang and Rakhlin [4]	Recovery	For $k/\sqrt{n} \to \infty$. Succeeds when $\mu \geq C(\sqrt{n}/k + \sqrt{\log n/k})$.	Polynomial time
Thresholded spectral projection	Cai, Liang and Rakhlin [4]	Recovery	For $k/\sqrt{n} \to 0$. Succeeds when $\mu/\sqrt{\log(n/k^2)} \to \infty$.	Polynomial time

The max and spectral rows give sufficient conditions. Here, C is a sufficiently large absolute constant. The exponent comparisons use strict inequalities, as explained in Section 3.

2 Detection

2.1 The Model and Testing Error

We begin with the standard Gaussian formulation of the problem. We observe a matrix $X \in \mathbb{R}^{n \times n}$ and consider the null hypothesis

$$H_0 \quad X_{ij} \overset{\text{iid}}{\sim} N(0,1)$$

against the alternative

$$H_1 \quad X_{ij} \overset{\text{ind}}{\sim} N\left(\mu \mathbb{1}_{\{i \in S,\, j \in T\}}, 1\right)$$

where $[n] = \{1, \ldots, n\}$ and $S, T \subseteq [n]$ are unknown subsets with $|S| = |T| = k$. The signal strength is $\mu > 0$, and $\mathbb{1}$ denotes the indicator function. Thus, under the alternative, there is an unknown $k \times k$ submatrix whose entries have mean μ, while all other entries have mean zero. The entries are independent under both hypotheses, and the affected rows and columns need not be consecutive.

Our goal is to understand how the difficulty of detection changes as the matrix becomes larger. As n increases, the block size and signal strength may also change, so we write $k = k_n$ and $\mu = \mu_n$. For instance, a larger block may allow us to detect a weaker signal because more affected entries contribute evidence for the alternative. We seek conditions on n, k, and μ under which the testing errors tend to zero. Throughout the asymptotic results, we assume $k \to \infty$ and $k = o(n)$.

Write P_0 for the null distribution and $P_{S,T,\mu}$ for the distribution under an alternative with affected rows S and columns T. A test φ_n takes value one when we reject H_0 and zero otherwise. We allow the test to depend on n, since its input and rejection threshold may change with the matrix size. Its total testing error is

$$R_n(\varphi_n) = P_0(\varphi_n = 1) + \sup_{\substack{S,T \subseteq [n] \\ |S| = |T| = k}} P_{S,T,\mu}(\varphi_n = 0)$$

The first term is the false positive rate. The second is the largest probability of missing a signal over all possible block locations. We say that reliable detection is achieved if $R_n(\varphi_n) \to 0$, so that both errors vanish. This is the minimax testing criterion used by Butucea and Ingster [1] and Ma and Wu [2], written in our notation.

2.2 The Sum Test

The first, and most natural (in fact, we both had thought of this before even looking at any papers), approach is to sum all entries in the matrix and compare the result with what is expected under H_0. As only detection is required, such a method is able to aggregate the effects of individually small mean shifts without identifying the specific indices. We define

$$T_{\text{sum}} = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{n} X_{ij}$$

This statistic is $N(0,1)$ under H_0 and $N(\Delta_n, 1)$ under any alternative. Here, $\Delta_n = \mu k^2/n$ is the mean shift of the normalized sum. We reject when $T_{\text{sum}} > c_n$, where c_n is the rejection threshold. Theorem 2.1 of Butucea and Ingster [1] shows that reliable detection is possible when $\Delta_n \to \infty$. We can then choose c_n to tend to infinity while staying below a fixed fraction of Δ_n. Their error calculation is given in Appendix A.1.

Computing the sum takes $O(n^2)$ operations, which is linear in the number of observed entries, and only $O(1)$ additional storage if the entries are processed sequentially. Its limitation is statistical in the sense that the unnormalized sum collects a mean shift of μk^2, but also noise with standard deviation n. For sufficiently small blocks, the many unaffected entries can obscure the signal. This motivates procedures that concentrate evidence within candidate submatrices instead.

2.3 The Scan Test

A natural alternative is to sum only over a candidate submatrix. Since the affected rows and columns are unknown, we consider all possible candidates and reject if any candidate has a sufficiently large sum. For each candidate with row set A and column set B, both of size k, let $Z_{A,B} = k^{-1} \sum_{i \in A, j \in B} X_{ij}$ be its normalized sum. There are $M_n = \binom{n}{k}^2$ candidates. The scan statistic is the largest of their normalized sums

$$T_{\text{scan}} = \max_{\substack{A,B \subseteq [n] \\ |A|=|B|=k}} Z_{A,B}$$

Each candidate sum is standard normal under H_0, while the true block satisfies $Z_{S,T} \sim N(\mu k, 1)$ under H_1. However, searching over many candidates makes it more likely that some sum will be large purely because of noise. The threshold must account for this search.

Let h_n denote the rejection threshold, so that we reject H_0 when $T_{\text{scan}} > h_n$. Butucea and Ingster [1, Theorem 2.1] choose $h_n = \sqrt{2 \log M_n}$. This threshold grows with the number of candidates to control false positives from the search. With this choice, they show that reliable detection is possible when

$$\liminf_{n \to \infty} \frac{\mu^2 k}{4 \log(n/k)} > 1$$

Thus, it is sufficient for the entrywise mean shift μ to exceed $2\sqrt{\log(n/k)/k}$ by a fixed factor. Appendix A.2 follows their proof, including the control of the maximum over dependent candidate sums.

The difficulty is computational. The scan requires knowledge of k, and exhaustive enumeration examines M_n candidates. Computing each sum separately takes $O(k^2 M_n)$ operations, which is superpolynomial under our assumptions on k. A better statistical guarantee can therefore come at a substantial computational cost.

2.4 The Maximum-Entry Test

Another natural approach is related to the sparse inference problems discussed in the course, where we considered minimum-p tests and Tukey's higher criticism. A minimum-p test looks

for a single unusually significant observation, while higher criticism can collect evidence from an excess of small p-values. In our setting, the first approach corresponds to testing whether any matrix entry is unusually large. Accordingly, define

$$T_{\max} = \max_{1 \le i,j \le n} X_{ij}$$

Let Φ denote the standard normal distribution function. Since the one-sided p-value for each entry is $p_{ij} = 1 - \Phi(X_{ij})$, we have $\min_{i,j} p_{ij} = 1 - \Phi(T_{\max})$. Thus, this is precisely a minimum-p test.

For this test, let u_n denote the rejection threshold, so that we reject when $T_{\max} > u_n$. Ma and Wu [2, Lemma 1] choose $u_n = \sqrt{(4+\varepsilon)\log n}$, where $\varepsilon > 0$ is a fixed constant. Their Theorem 2, condition (17), gives the sufficient regime

$$\liminf_{n\to\infty} \frac{\mu}{2\sqrt{\log n}} > 1$$

Taking ε sufficiently small then makes the total error tend to zero. The bound they use is stated and explained in Appendix A.3.

Like the sum, the maximum can be computed in $O(n^2)$ operations with $O(1)$ additional storage. It requires neither the block size nor its location. However, it does not exploit the shared row and column structure, since permuting all matrix entries leaves the statistic unchanged.

3 Comparison

3.1 Statistical Detection Regimes

The sum and scan tests differ in how they use the affected entries. The sum gains from having many small shifts across the matrix. The scan avoids much of the noise from unaffected entries, but must compensate for the number of possible block locations. Butucea and Ingster [1, Theorem 2.1] combine the two tests, rejecting whenever either rejects, so that the combined procedure succeeds under either of the conditions above.

Their lower bound shows why both conditions matter. Write $R_n^\star = \inf_\varphi R_n(\varphi)$, where the infimum is over all tests based on the full matrix. In the square case, their Theorem 2.2 gives

$$\frac{\mu k^2}{n} \longrightarrow 0 \qquad \text{and} \qquad \limsup_{n\to\infty} \frac{\mu^2 k}{4\log(n/k)} < 1 \quad \Longrightarrow \quad R_n^\star \longrightarrow 1$$

The additional balance assumptions in the theorem hold in this square setting, and its proof uses alternatives with a common mean shift [1, Section 5.1]. Thus, when both ways of accumulating signal are too weak, no other test can make the errors vanish. The strict inequalities do not settle all critical sequences at equality.

To see which test is useful for different block sizes, follow Ma and Wu [2, Section 1.2] and write $k = \lfloor n^\gamma \rfloor$ and $\mu = n^{-\beta}$, with $0 < \gamma < 1$ and $0 < \beta < 1$. Larger β means a weaker

signal. Their statistical boundary, in our notation, is

$$\beta_{\text{stat}} = \max\left\{\frac{\gamma}{2},\, 2\gamma - 1\right\}$$

Detection is statistically possible for $\beta < \beta_{\text{stat}}$ and impossible for $\beta > \beta_{\text{stat}}$.

For $2/3 < \gamma < 1$, which Ma and Wu call the moderately sparse regime, the sum attains the statistical rate. The block contributes enough total signal that searching for its location does not improve this rate. For $0 < \gamma < 2/3$, their highly sparse regime, scanning attains the statistical boundary at weaker signals than the sum. At $\gamma = 2/3$, the two exponent expressions agree. The original conditions must still be used to account for logarithmic factors.

The max guarantee requires individual mean shifts larger than $2\sqrt{\log n}$ by a fixed factor. These signals are stronger than those described by $\mu = n^{-\beta}$. The max is inexpensive and useful when individual entries are informative, but Ma and Wu's stated sufficient bound does not give a sharp classification of its power for every block size.

3.2 Computation

Both the sum and max take time linear in the number of entries. The statistically stronger scan guarantee for small blocks is obtained by a search whose direct implementation is superpolynomial. This alone does not prove that efficient detection is impossible, since another algorithm might avoid the search.

Ma and Wu [2, Theorem 2 and Figure 1] address this question under the planted-clique hypothesis. This hypothesis rules out reliable randomized polynomial-time detection of a planted clique of size N^{ρ}, for fixed $0 < \rho < 1/2$, in a graph drawn from $G(N, 1/2)$. Their reduction shows that an efficient submatrix detector in certain regimes would also give an efficient detector for such a clique.

Ma and Wu allow a rectangular affected block with at least k rows and columns and means bounded below by μ. They formulate the computational theorem for a discretized Gaussian model that is statistically equivalent to the continuous model asymptotically. Their impossibility statement is for that larger alternative class. It does not automatically become an impossibility theorem for our exact-size, common-mean alternative.

For their model, the computational exponent boundary is $\beta_{\text{comp}} = \max\{0, 2\gamma - 1\}$. When $0 < \gamma < 2/3$, the interval $\beta_{\text{comp}} < \beta < \beta_{\text{stat}}$ is statistically possible but computationally intractable under that hypothesis. When $2/3 < \gamma < 1$, the two boundaries coincide and the sum already achieves the statistical rate. These are strict exponent comparisons. Logarithmic factors can matter at the boundaries.

These asymptotic results do not determine the finite-sample power ordering of the tests. Our simulations will compare power at a common significance level and record the computational cost. Any heuristic approximation to the scan will be distinguished from the exhaustive statistic. Butucea and Ingster also use a heuristic search in their numerical study [1, Section 3].

4 Recovery

Detection establishes that an affected group exists, but does not tell us which rows and columns belong to it. Recovering these indices may be necessary if we want to investigate the group further. A recovery procedure returns estimates $\widehat{S}$ of the affected rows and $\widehat{T}$ of the affected columns. We require

$$\sup_{\substack{S,T\subseteq[n]\\|S|=|T|=k}} P_{S,T,\mu}\big((\widehat{S},\widehat{T}) \neq (S,T)\big) \longrightarrow 0$$

This is exact recovery. Both index sets must be entirely correct, so even one incorrect row or column counts as an error.

4.1 Exact Recovery by Scanning

Butucea, Ingster and Suslina [3] study this problem using the estimator

$$(\widehat{S},\widehat{T}) = \arg\max_{\substack{A,B\subseteq[n]\\|A|=|B|=k}} \sum_{i\in A, j\in B} X_{ij}$$

This is the same search used by the scan test, but now we retain the maximizing indices. The true block must now have a larger sum than every incorrect candidate, including candidates that differ by just one row or column.

For $k \to \infty$ and $k = o(n)$, their Theorems 2.1 and 2.3 give the sharp square-block recovery boundary

$$\mu^\star_{\mathrm{rec}} \sim \max\left\{ \frac{\sqrt{2\log k} + \sqrt{2\log(n-k)}}{\sqrt{k}},\ 2\sqrt{\frac{\log(n/k)}{k}} \right\}$$

A fixed multiplicative separation above this boundary permits exact recovery, while a fixed separation below it makes the minimax probability of an incorrect recovery tend to one.

The first term explains a difficulty that detection does not have. Even if the true columns were known, the affected rows would have to be separated from every unaffected row. The standardized row sums have mean shift $\mu\sqrt{k}$, which must overcome both unusually large noise rows and unusually small affected rows. The second term accounts for the many competing block locations.

The exhaustive estimator has the same computational difficulty as the scan test. We next consider an efficient alternative.

4.2 Spectral Recovery

Cai, Liang and Rakhlin [4] approach recovery using the singular vectors of the observed matrix. The mean matrix is $M = \mu \mathbb{1}_S \mathbb{1}_T^\top$, which has rank one. Here, $\mathbb{1}_S$ and $\mathbb{1}_T$ are the indicator vectors of the affected rows and columns. The left and right singular vectors of M are proportional to these vectors.

Their Algorithm 1 computes the leading left singular vector $\widehat{u}$ and leading right singular vector $\widehat{v}$ of X. It then forms the column scores $\widehat{u}^\top X_{\cdot j}$ and row scores $X_{i\cdot}\widehat{v}$. Under their signal condition, the affected and unaffected indices form two separated groups of scores, and a threshold between the groups recovers the index sets.

For square blocks with $k/\sqrt{n} \to \infty$, their Lemma 1 gives a sufficiently large absolute constant C such that exact recovery succeeds whenever

$$\mu \geq C\left(\frac{\sqrt{n}}{k} + \sqrt{\frac{\log n}{k}}\right)$$

The authors suggest iterative power methods for the leading singular vectors, avoiding the search over block locations.

4.3 Smaller Blocks and Computational Limits

For $k/\sqrt{n} \to 0$, Cai, Liang and Rakhlin first reduce the noise by soft-thresholding each entry. At a threshold level $t > 0$, this replaces an entry x by $\eta_t(x) = \mathrm{sign}(x)(|x| - t)_+$, where $(a)_+ = \max\{a, 0\}$. They then apply the same spectral projection and clustering steps to the thresholded matrix. Their Theorem 4 gives exact recovery when

$$\frac{\mu}{\sqrt{\log(n/k^2)}} \longrightarrow \infty$$

with a threshold level t of order $\sqrt{\log(n/k^2)}$.

Under a hidden-clique recovery hypothesis, their Theorems 2–4 give a gap between statistical and computational recovery in the broader sub-Gaussian model, even in dense regimes where the detection gap disappears. The spectral upper bounds apply to our Gaussian observations, while the computational lower bounds are minimax statements over the paper's larger noise class.

5 Simulations

We used three fixed simulation designs, summarized in Table 2. Under the null, the entries of each $n \times n$ matrix were independent standard Gaussians. Under the alternative, we sampled k rows and k columns uniformly without replacement and added μ to their intersections. These indices were unknown to the procedures. All results in this section use 1,500 independent null calibration matrices per configuration and 600 fresh null and 600 fresh planted detection matrices per signal. Studies with recovery use another 600 planted matrices per signal. Calibration data are separate from evaluation data; procedures share the same matrix within each evaluation trial.

Table 2. Final simulation designs. Evaluation counts exclude calibration matrices.

Study	Parameter grid	Pairs	Evaluations
Broad sweep	Seven matrix sizes; five requested sparsity exponents; five scan-scaled signals	31	279,000
Signal references	Four matrix/block pairs; half, once, and twice each applicable signal reference, plus zero	4	126,000
Power laws	Three matrix sizes; four requested exponents; beta values at and around both detection boundaries	12	172,800

For the broad sweep, $n \in \{6, 8, 12, 16, 24, 32, 64\}$ and requested $\gamma \in \{.25, .40, .55, .70, .85\}$. We set $k = \lfloor n^{\gamma} \rfloor$, clipped to $[2, n-1]$, and $\mu = c\, 2\sqrt{\log(n/k)/k}$ for $c \in \{0, .65, 1, 1.35, 1.8\}$. Repeated integer block sizes are simulated once; the effective exponent is $\log k / \log n$. The reference study uses $(n, k) = (12, 2), (12, 5), (64, 4), (64, 16)$ and samples both sides of every applicable sum, scan, maximum-entry, exact-recovery, spectral and thresholded-spectral signal reference from the theory sections. A reference multiplier of one is not a finite-sample cutoff: the sum condition and the small-block spectral condition require diverging ratios, and the large-block spectral guarantee has an unspecified sufficiently large constant.

In the power-law study, $n \in \{12, 24, 64\}$, requested $\gamma \in \{.4, .5, 2/3, .8\}$, and $\mu = n^{-\beta}$. We include $\beta \in \{0, .1, .2, .3, .4, .6, .8\}$, both exponent boundaries computed from the effective exponent, and offsets $\pm .04$ where nonnegative. A separate $\mu = 0$ evaluation is included; $\beta = 0$ instead means $\mu = 1$.

5.1 Calibration and finite-sample error checks

Detection uses nominal level .05 and an upper order-statistic cutoff $\lceil (m+1)(1-\alpha) \rceil$ from $m = 1{,}500$ calibration draws. Sum-or-scan uses two level-.025 cutoffs. Wilson 95% intervals describe evaluation uncertainty conditional on the cutoff and do not include calibration variability. Exhaustive scanning and block maximization are computed only when $\binom{n}{k} \leq 2{,}500$; uncomputed methods are omitted. Spectral methods use a full SVD and the known k. Small-block recovery also uses the fixed soft threshold $t = \sqrt{2 \log(n/k^2)}$.

All 577,800 evaluation records were checked against their summary counts. We compared 1,476 sum and maximum-entry rejection rates with their exact Gaussian probabilities. A

separate check reused the reference study's held-out matrices with $c_n = \mu k^2/(2n)$, $h_n = \sqrt{2\log\binom{n}{k}^2}$ and $u_n = \sqrt{4.5\log n}$ from the error calculations. We checked sum errors against their exact Gaussian probabilities, scan errors against the union and true-block bounds, and maximum-entry errors against the stated bounds. All 350 bound checks were consistent. The exact-probability and error-bound comparisons each use exact binomial checks with a Bonferroni-adjusted 1% familywise level. These diagnostics assess finite-sample implementation consistency; they do not verify asymptotic impossibility or computational-hardness claims.

5.2 Results across regimes

Table 3 gives selected points from the reference study. Figures 1 and 2 show all sampled signals for its four configurations. The small-matrix cases allow direct comparison with exact search. The larger cases compare efficient methods in small- and large-block recovery regimes. Stronger signals yield high exact-recovery probabilities, while the signal needed for exact recovery can exceed that needed for detection. The full broad sweep, null checks, timings and raw trials are available in the executed notebook and repository.

Table 3. Selected final-study estimates, with Wilson 95% intervals from 600 trials.

(n, k)	μ	Procedure / outcome	Estimate [95% CI]
$(64, 16)$	0.250	Global sum / power	0.253 $[0.220, 0.290]$
$(64, 16)$	0.500	Global sum / power	0.608 $[0.569, 0.647]$
$(64, 4)$	4.079	Maximum entry / power	1.000 $[0.994, 1.000]$
$(12, 2)$	2.350	Block maximization / recovery	0.723 $[0.686, 0.758]$
$(12, 2)$	4.700	Block maximization / recovery	1.000 $[0.994, 1.000]$
$(64, 4)$	3.020	Spectral / recovery	0.775 $[0.740, 0.807]$
$(64, 4)$	3.020	Thresholded spectral / recovery	0.913 $[0.888, 0.933]$
$(64, 16)$	2.020	Spectral / recovery	1.000 $[0.994, 1.000]$

Figure 3 displays the power-law study. The exponent lines are asymptotic references, with logarithmic factors omitted; they do not predict a sharp transition at these finite dimensions. In particular, the smallest matrices are diagnostics rather than evidence of a limiting theorem. The original D1–D4 and R1–R4 results have been superseded in this section; their historical source and data remain available in repository history.

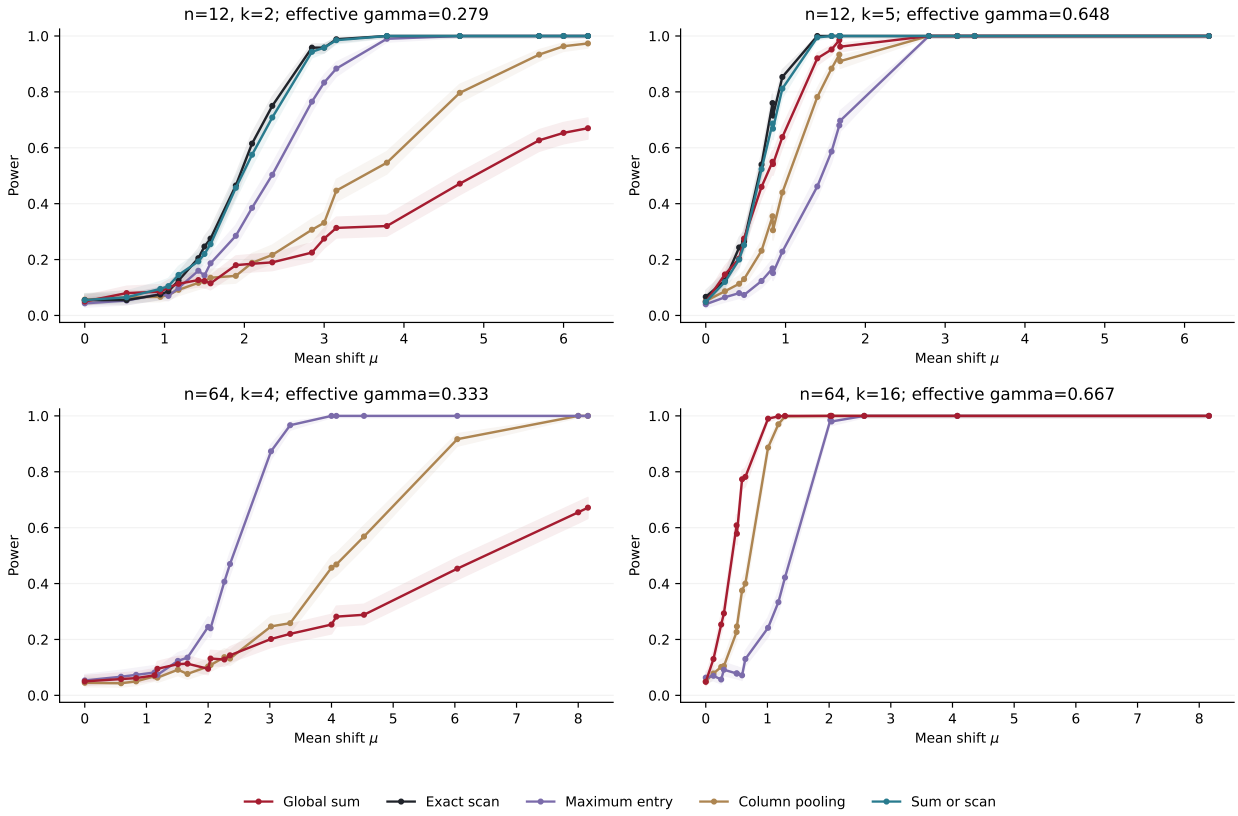


Figure 1. Detection power in the signal-reference study. All sampled mean shifts are shown; shaded bands are marginal Wilson 95% intervals. Exact scanning and the combined test are present only in feasible small-matrix cases.

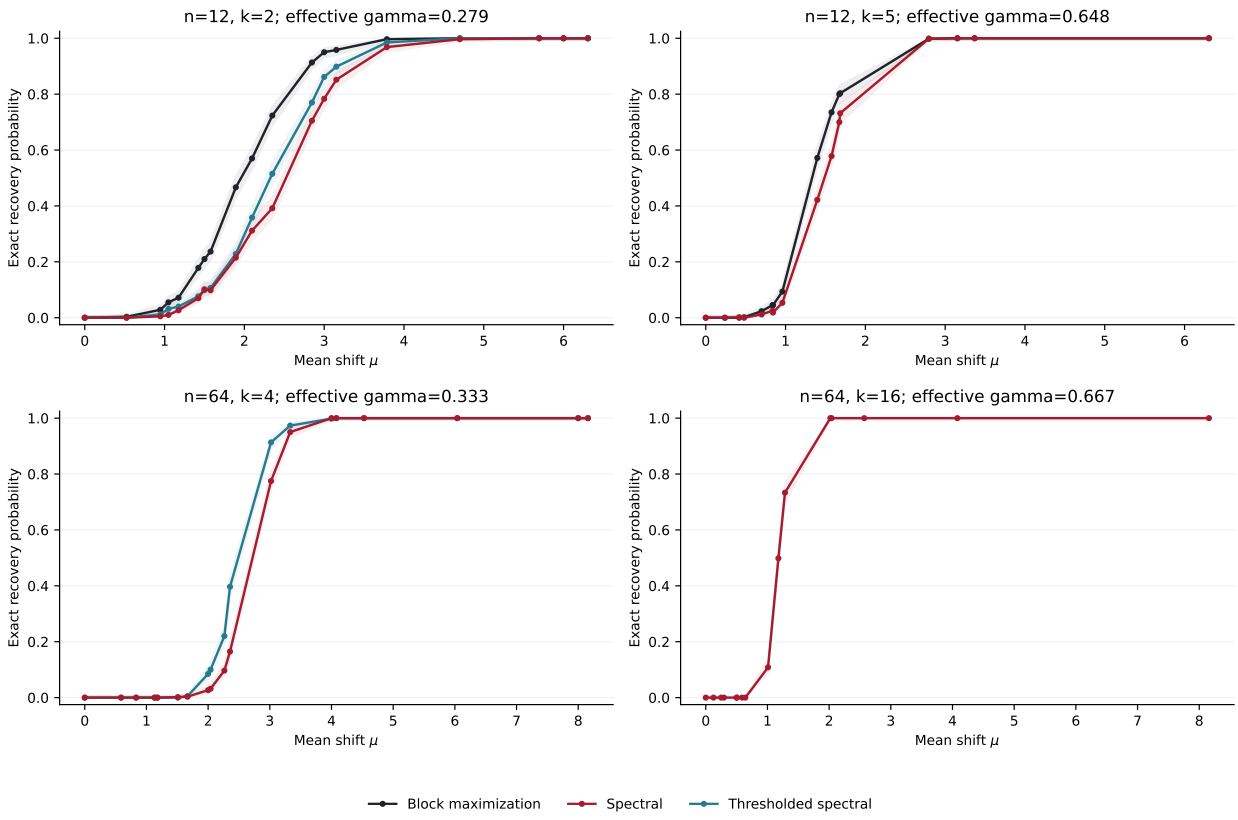


Figure 2. Exact recovery in the signal-reference study. Both planted index sets must be correct. The thresholded spectral method is included only for $k^2 < n$.

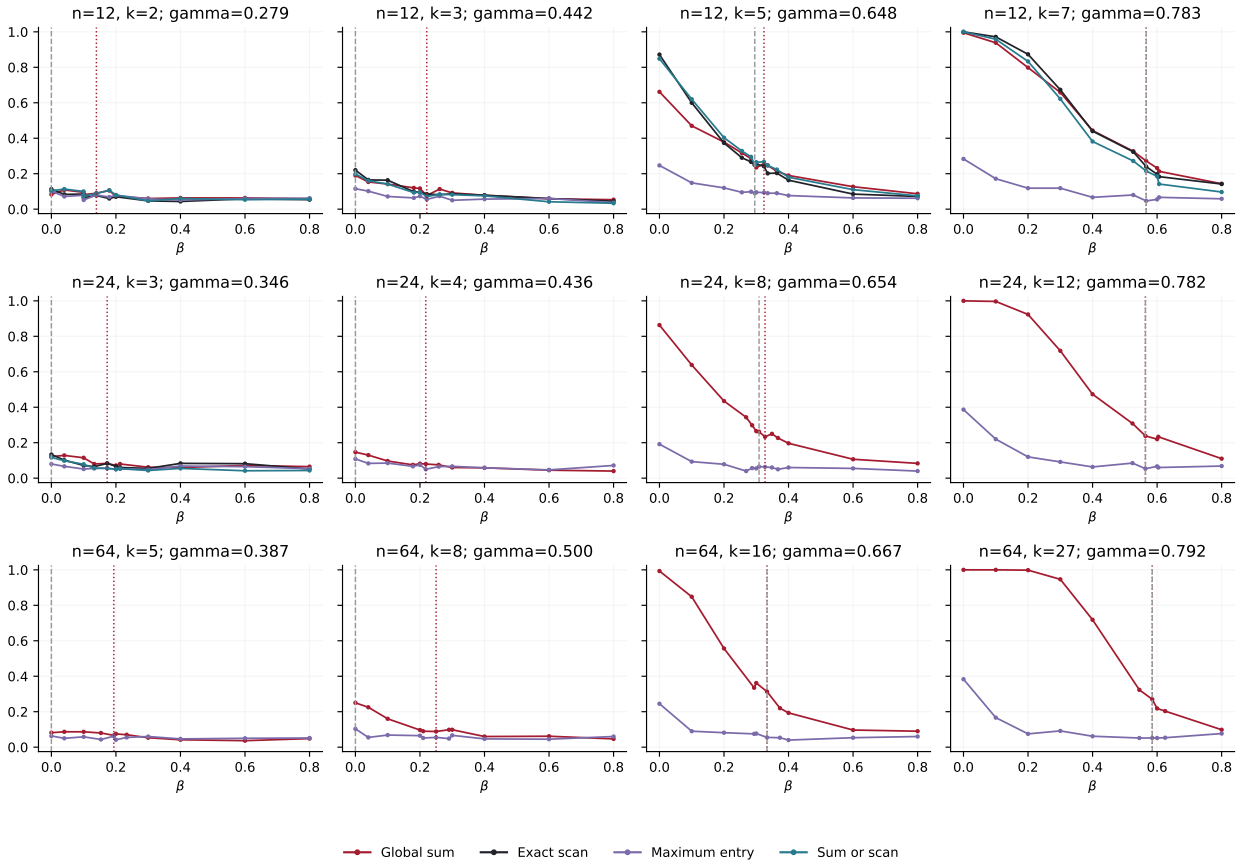


Figure 3. Detection across the power-law study. Larger β gives weaker signals. Dotted red and dashed gray lines are the statistical and conditional computational exponent references for the actual rounded block size. Exact scanning is omitted when the row-subset cap is exceeded.

A Detection Calculations

The sum and scan calculations follow Appendix A.1 of Butucea and Ingster [1]. Their large matrix dimensions N, M are both n here, their submatrix dimensions are both k, and their parameter a is our μ. For Ma and Wu [2], the change of notation is $p = n$ and $\lambda = \mu$.

A.1 The Sum Test

Since the entries are independent and each has variance one, the unnormalized sum has variance n^2 under both hypotheses. Dividing by n therefore gives T_{sum} variance one. Under H_1, exactly k^2 entries have mean μ, so the unnormalized sum has mean μk^2. Moreover, a linear combination of independent Gaussian variables is Gaussian. Consequently,

$$T_{\text{sum}} \sim \begin{cases} N(0,1), & H_0 \\ N(\Delta_n, 1), & H_1 \end{cases} \qquad \Delta_n = \frac{\mu k^2}{n}$$

Let c_n denote the rejection threshold, so that we reject H_0 whenever $T_{\text{sum}} > c_n$. The two errors are

$$P_0(T_{\text{sum}} > c_n) = \Phi(-c_n) \qquad \sup_{S,T} P_{S,T,\mu}(T_{\text{sum}} \le c_n) = \Phi(c_n - \Delta_n)$$

These are the expressions used in the paper's proof. Its upper bound on the Type II error is an equality in our common-mean model.

Following their threshold choice, suppose $\Delta_n \to \infty$ and take $c_n \to \infty$ with $c_n \le \eta \Delta_n$ for a fixed $0 < \eta < 1$. Then the Type I error tends to zero and

$$\sup_{S,T} P_{S,T,\mu}(T_{\text{sum}} \le c_n) \le \Phi\big(-(1-\eta)\Delta_n\big) \longrightarrow 0$$

Thus, the sum test gives reliable detection when $\mu k^2 / n \to \infty$.

A.2 The Scan Test

For fixed A, B with $|A| = |B| = k$, the unnormalized candidate sum has variance k^2. Hence $Z_{A,B} \sim N(0,1)$ under H_0. Under an alternative, the true candidate satisfies $Z_{S,T} \sim N(\mu k, 1)$.

For any proposed rejection threshold $u > 0$, a union bound gives

$$P_0(T_{\text{scan}} > u) \le \sum_{\substack{A,B \subseteq [n] \\ |A|=|B|=k}} P_0(Z_{A,B} > u) = M_n \Phi(-u)$$

This does not require independence between candidate sums, which is important because overlapping submatrices share observations.

Set the rejection threshold to $h_n = \sqrt{2 \log M_n}$, so that we reject when $T_{\text{scan}} > h_n$. Butucea and Ingster use the Gaussian tail estimate

$$\Phi(-u) \asymp \frac{e^{-u^2/2}}{u} \qquad \text{as } u \to \infty$$

Here, $\asymp$ means that the quantities are bounded above and below by fixed positive multiples of one another for sufficiently large u. Since $M_n \to \infty$, we have $h_n \to \infty$, and their argument gives

$$P_0(T_{\text{scan}} > h_n) \le M_n \Phi(-h_n) = O\left(\frac{M_n e^{-h_n^2/2}}{h_n}\right) = O\left(\frac{1}{h_n}\right) \longrightarrow 0$$

The final equality follows from $h_n^2/2 = \log M_n$. The factor $1/h_n$ makes the error vanish at this threshold.

We next control the Type II error. Under any alternative, the maximum includes the true block, so $T_{\text{scan}} \ge Z_{S,T}$. Consequently,

$$\begin{aligned} P_{S,T,\mu}(T_{\text{scan}} \le h_n) &\le P_{S,T,\mu}(Z_{S,T} \le h_n) \\ &= \Phi(h_n - \mu k) \end{aligned}$$

The bound is independent of S, T, so it also holds for the worst-case Type II error.

Equation (A.1) of the paper gives, in the square case,

$$\log M_n \sim 2k \log(n/k) \qquad h_n \sim 2\sqrt{k \log(n/k)}$$

The condition $\liminf \mu^2 k/[4 \log(n/k)] > 1$ therefore gives a fixed multiplicative separation between μk and h_n. Since $h_n \to \infty$, their difference tends to infinity, so the Type II error also vanishes. This is the square specialization of condition (2.5) in their Theorem 2.1.

A.3 The Maximum-Entry Test

Lemma 1, equation (12), of Ma and Wu [2] states that, for $u_n = \sqrt{(4+\varepsilon)\log n}$ with fixed $\varepsilon > 0$,

$$R_n\big(\mathbb{1}_{\{T_{\max} > u_n\}}\big) \le n^{-\varepsilon/2} + \exp\left[-\frac{1}{2}(\mu - u_n)_+^2\right]$$

Here, $x_+ = \max\{x, 0\}$. We follow their argument in Section 6.1. Under H_0, a union bound and the Gaussian Chernoff bound used there give

$$P_0(T_{\max} > u_n) \le n^2 \Phi(-u_n) \le n^2 e^{-u_n^2/2} = n^{-\varepsilon/2}$$

Under H_1, choose an affected entry $X_{ij} \sim N(\mu, 1)$. Since $T_{\max} \ge X_{ij}$,

$$P_{S,T,\mu}(T_{\max} \le u_n) \le P_{S,T,\mu}(X_{ij} \le u_n) = \Phi(u_n - \mu) \le \exp\left[-\frac{1}{2}(\mu - u_n)_+^2\right]$$

This holds for every location and gives the stated total error bound. In Section 4.3, the authors apply it under the second condition in (17) of Theorem 2. If $\liminf \mu/(2\sqrt{\log n}) > 1$, taking ε sufficiently small makes $\liminf \mu/u_n > 1$, and both error bounds tend to zero.

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